

ASSIGNMENT 1

1. Difference between Frontend, Backend, and Full-Stack Development

- **Frontend:**

It's what users **see and interact with** on a website.

Example: Buttons, text boxes, colors, menus on Facebook or Amazon.

Technologies: **HTML, CSS, JavaScript**

- **Backend:**

It's what works **behind the scenes** — manages data, logic, and servers.

Example: When you log in to Instagram, the backend checks your username and password.

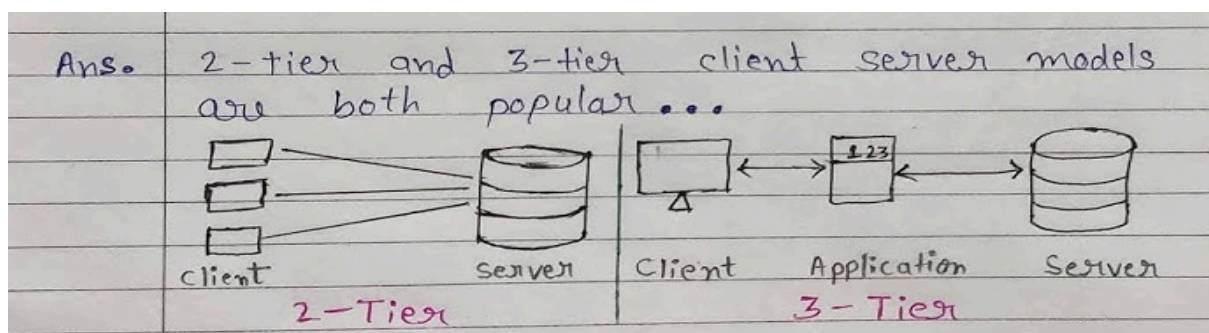
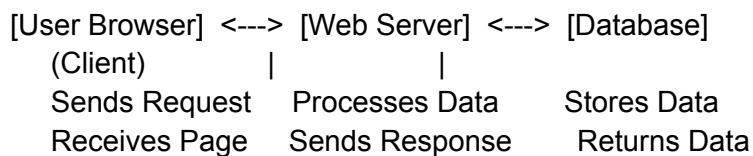
Technologies: **Python, Node.js, PHP, Java**

- **Full-Stack:**

A developer who works on **both frontend and backend**.

Example: A developer who builds both the login form (frontend) and the login system (backend).

2. Simple Diagram: Client-Server Model



3. How a Browser Requests and Displays a Web Page

1. You type a web address (like www.google.com) in the browser.
 2. The browser sends a **request** to the **web server**.
 3. The server **finds the requested page** (HTML, CSS, JS files).
 4. The server **sends those files** back to the browser.
 5. The browser **renders (displays)** the web page on your screen.
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4. Tools for Web Development Environment

Tool	Purpose
VS Code	Writing and editing code
Web Browser (Chrome, Firefox)	Testing and viewing websites
Node.js	Running JavaScript outside the browser
Git & GitHub	Version control and project sharing
Live Server Extension (in VS Code)	Automatically refreshes the page when code changes

5. What is a Web Server?

A **web server** is software or hardware that **stores websites** and **delivers web pages** to users when they request them.

Examples:

- Apache
 - Nginx
 - Microsoft IIS
 - LiteSpeed
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6. Roles in a Project

- **Frontend Developer:** Designs and builds the user interface (UI).
 - **Backend Developer:** Manages server-side logic, APIs, and database connections.
 - **Database Administrator (DBA):** Maintains the database — manages data, backups, and performance.
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7. Installing and Configuring VS Code (HTML, CSS, JS)

Steps:

1. Download and install **VS Code** from <https://code.visualstudio.com>.
 2. Install extensions:
 - *Live Server* (for previewing webpages)
 - *Prettier* (for code formatting)
 - *HTML CSS Support*
 3. Create a folder → open in VS Code → create files:
 - `index.html`
 - `style.css`
 - `script.js`
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8. Static vs Dynamic Websites

Type	Description	Example
Static Website	Same content for every user; built with HTML & CSS only	A personal portfolio website
Dynamic Website	Content changes based on user actions or data	Facebook, YouTube, Amazon

9. Five Web Browsers and Their Rendering Engines

Browser	Rendering Engine
Google Chrome	Blink
Mozilla Firefox	Gecko
Microsoft Edge	Blink
Apple Safari	WebKit
Opera	Blink

Rendering engine: Converts HTML, CSS, and JS into what you visually see on screen.
Different engines = Slight differences in how pages look or behave.

10. Basic Web Architecture Flow

